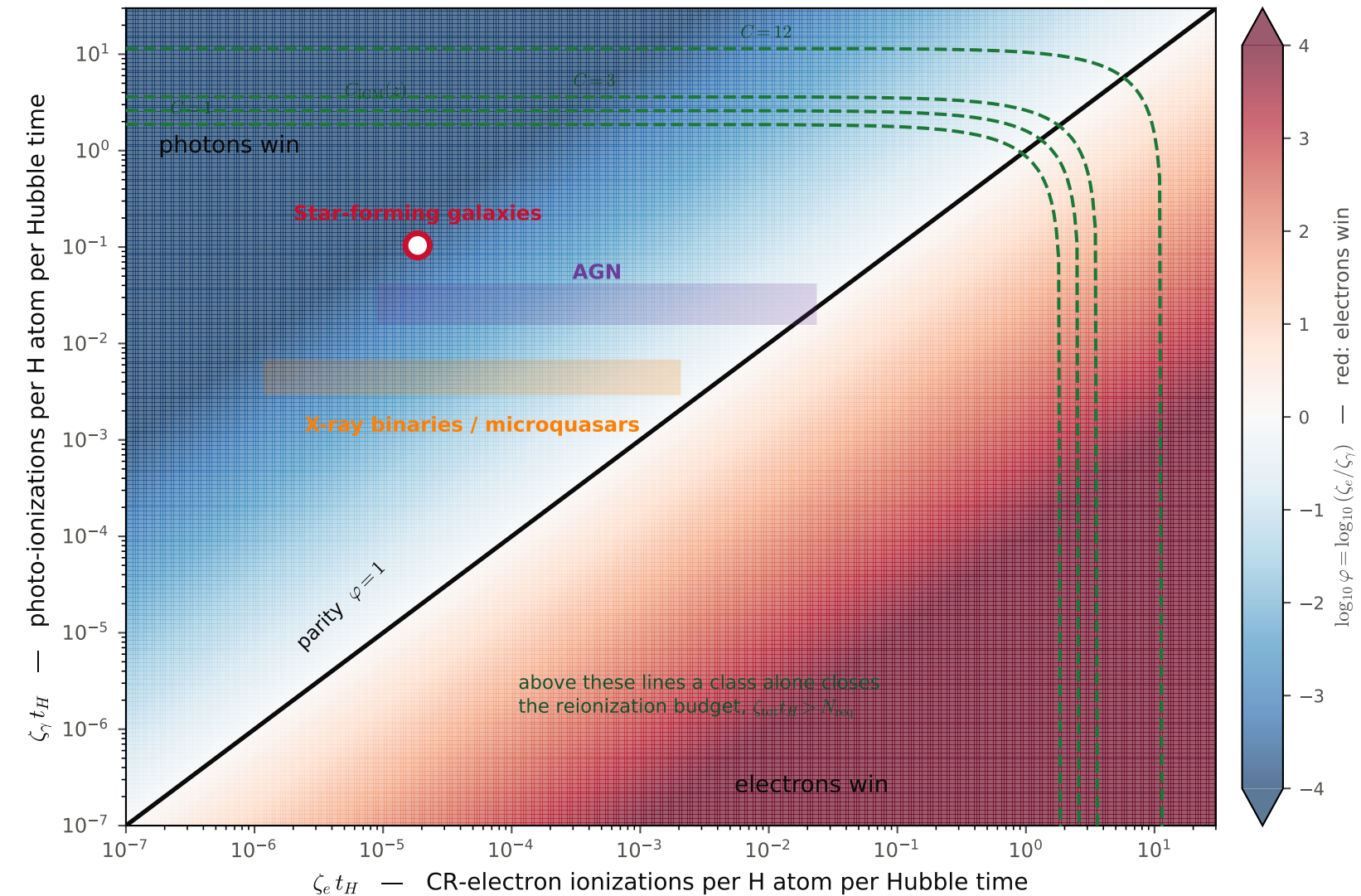


Which high- z sources favour CR electrons? Comoving populations at $z = 10$



IGM at $z=10$, $x_e=10^{-4}$, pure H; $t_H=1/H(z)$ from astropy Planck18 = 0.7086 Gyr.

Axes carry BOTH luminosity and spectrum: $\zeta=CL/(n_H W)$, verified against the full integral (check S2). Photon yields carry the IGM transparency factor $1-e^{-\tau(E)}$.

Star-forming galaxies are a POINT (both luminosities are the project's own anchored chain, reproducing $\zeta_\gamma/\zeta_e=5599$). AGN and X-ray binaries are BOXES: their L_e spans the scanned 10^{-3} - 10^0 jet electron fraction, since the proton content of jets is not observationally constrained.

AGN L_{ion} : 17-33% of the LyC budget (Asthana+24; Jiang+25 upper bound), measured at $z\sim 7.5$ and carried to $z=10$ — conservative, the AGN LF falls faster than the galaxy LF. XRB L_X : Lehmer+16 eq.(6), carried to the ionizing band with each class's own sourced spectrum.

$f_{\text{dep}}=1$ (assumption A11): every injected electron is assumed to deposit, so every ζ_e here is a CEILING.